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PROVISIONAL PATENT APPLICATION

TITLE OF INVENTION

Multi-Sided Regulatory Infrastructure System with AI-Driven Cross-Module Routing and Unified Access Control

INVENTOR

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FIELD OF THE INVENTION

The present invention relates generally to computer-implemented systems for regulatory management, and more specifically to a **multi-domain, modular infrastructure platform** integrating healthcare, legal, compliance, and business systems through artificial intelligence, automated workflows, and unified access control.

BACKGROUND OF THE INVENTION

Modern regulatory environments are fragmented across multiple independent systems, including healthcare platforms, legal services, compliance systems, and business licensing infrastructures. These systems operate in isolation, resulting in inefficiencies, duplicated data entry, lack of interoperability, and increased risk of non-compliance.

Existing solutions fail to provide a unified system capable of coordinating activities across multiple regulated domains in real time. Additionally, current platforms do not effectively integrate artificial intelligence for intake, workflow automation, and cross-domain routing.

There is a need for a centralized system that integrates multiple regulated domains into a single infrastructure while maintaining compliance, security, and scalability.

SUMMARY OF THE INVENTION

The present invention provides a **multi-sided regulatory infrastructure system** that integrates multiple domain-specific modules within a unified platform.

The system includes:

- a centralized operating system
- modular components for healthcare, legal, compliance, and business operations
- a cross-module routing engine
- an artificial intelligence intake and processing system
- a unified role-based access control (RBAC) system
- a compliance automation engine
- and an audit and monitoring system

The invention enables users to interact with multiple regulatory domains through a single interface, while automatically routing workflows, enforcing compliance, and generating required documentation.

DETAILED DESCRIPTION OF THE INVENTION

1. System Architecture

The system operates as a centralized platform consisting of interconnected modules, including but not limited to:

- Healthcare Module (telehealth, patient management, physician systems)
- Legal Module (legal intake, attorney matching, case management)
- Compliance Module (regulatory validation, licensing, monitoring)
- Business Module (applications, licensing, operational management)

All modules operate within a shared data environment and are connected through a routing engine.

2. Cross-Module Routing Engine

The platform includes a routing engine configured to:

- transfer data and workflows between modules
- trigger actions across modules based on user activity
- respond to system conditions and regulatory requirements

Example:

A user interacting within a healthcare module may trigger:

- compliance validation
 - legal case creation
 - or business application processing
-

3. Artificial Intelligence Intake System

The system includes an artificial intelligence agent configured to:

- collect user input dynamically
- adjust questioning based on user responses
- structure data into standardized formats
- generate completed applications and documents
- assist users in completing regulatory, medical, and legal processes

The AI system functions as an automated intake and workflow engine.

4. Unified Role-Based Access Control (RBAC)

The system includes a unified RBAC system that:

- governs access across all modules
- assigns permissions based on user roles
- restricts access to sensitive information
- controls user actions within the system

Roles may include:

- patient
- physician
- attorney
- business operator
- administrator

5. Compliance Automation Engine

The system includes a compliance engine configured to:

- validate user input against regulatory requirements
 - detect missing or incorrect data
 - generate compliance-ready documentation
 - automate submission to regulatory bodies
 - track expiration and renewal requirements
-

6. Audit and Monitoring System

All system actions are:

- recorded in audit logs
- timestamped
- associated with user identity

The system maintains traceability across all modules.

7. Multi-Domain Data Integration

The system allows data generated in one module to be:

- shared with other modules
 - used to trigger workflows
 - integrated into a centralized data environment
-

8. Event-Driven System Behavior

System actions are triggered by:

- user input
- system events
- regulatory updates
- workflow completion

These events initiate automated processes across modules.

9. Regulatory Decision Support

The system provides:

- compliance recommendations
 - risk identification
 - decision support for approvals and denials
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10. Scalable Modular Infrastructure

The system supports:

- addition of new modules
 - integration of future technologies
 - expansion into additional regulated industries
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11. Cross-Domain User Interaction

Users may:

- interact with multiple modules
 - maintain a unified identity
 - transition between domains seamlessly
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12. Security Framework

The system includes:

- multi-factor authentication
- encrypted data storage

- access control enforcement
 - system-wide security monitoring
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13. Automated Document Generation

User input is transformed into:

- regulatory applications
 - compliance documents
 - structured data outputs
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14. Hybrid Human-AI Operation

The system combines:

- artificial intelligence automation
- human interaction
- real-time system enforcement

The system may be implemented using various computing environments, including cloud-based, distributed, or on-premise architectures. The components described may be combined, separated, or modified without departing from the scope of the invention.

CLAIMS

CLAIM 1 — CORE MULTI-SIDED REGULATORY SYSTEM

A computer-implemented system comprising:

- a centralized operating platform configured to manage data, users, and system operations;
- a plurality of domain-specific modules including healthcare, legal, regulatory compliance, and business operations;
- a shared data environment;
- a routing engine;
- and a role-based access control system

wherein the system enables real-time coordination across multiple regulated domains.

CLAIM 2 — CROSS-MODULE ROUTING ENGINE

The system of Claim 1, wherein the routing engine:

- directs users and workflows between modules
 - determines routing based on user role, system rules, regulatory requirements, or AI analysis
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CLAIM 3 — AI INTAKE SYSTEM

The system includes an artificial intelligence agent configured to:

- collect and structure user input
 - dynamically adjust questioning
 - generate application-ready outputs
-

CLAIM 4 — UNIFIED RBAC

A system wherein role-based access control governs permissions across all modules simultaneously.

CLAIM 5 — COMPLIANCE AUTOMATION

A system wherein compliance validation, document generation, and regulatory submission are automated.

CLAIM 6 — AUDIT SYSTEM

All system actions are recorded, timestamped, and traceable.

CLAIM 7 — DATA INTEGRATION

Data is shared and utilized across modules within a centralized environment.

CLAIM 8 — EVENT-DRIVEN PROCESSING

System actions are triggered by user input, system events, or regulatory changes.

CLAIM 9 — DECISION SUPPORT

The system provides compliance recommendations and regulatory decision support.

CLAIM 10 — SCALABILITY

The system supports modular expansion into additional domains.

CLAIM 11 — CROSS-DOMAIN USER FLOW

Users interact across multiple modules with a unified identity.

CLAIM 12 — AI WORKFLOW GENERATION

Artificial intelligence generates and manages workflows dynamically.

CLAIM 13 — SECURITY SYSTEM

The platform enforces authentication, encryption, and access controls.

CLAIM 14 — DOCUMENT AUTOMATION

User input is converted into completed regulatory documents.

CLAIM 15 — HYBRID SYSTEM

The system integrates AI processes with human oversight.

CLAIM 16 — CROSS-DOMAIN AUTOMATED CASE CREATION

A system wherein actions initiated in one module automatically generate corresponding records, workflows, or cases in a separate module without requiring additional user input.

ABSTRACT

A multi-sided regulatory infrastructure system integrating healthcare, legal, compliance, and business modules within a unified platform. The system includes artificial intelligence intake, cross-module routing, role-based access control, compliance automation, and audit tracking, enabling coordinated operation across multiple regulated domains.

FIGURE 1 — OVERALL SYSTEM ARCHITECTURE

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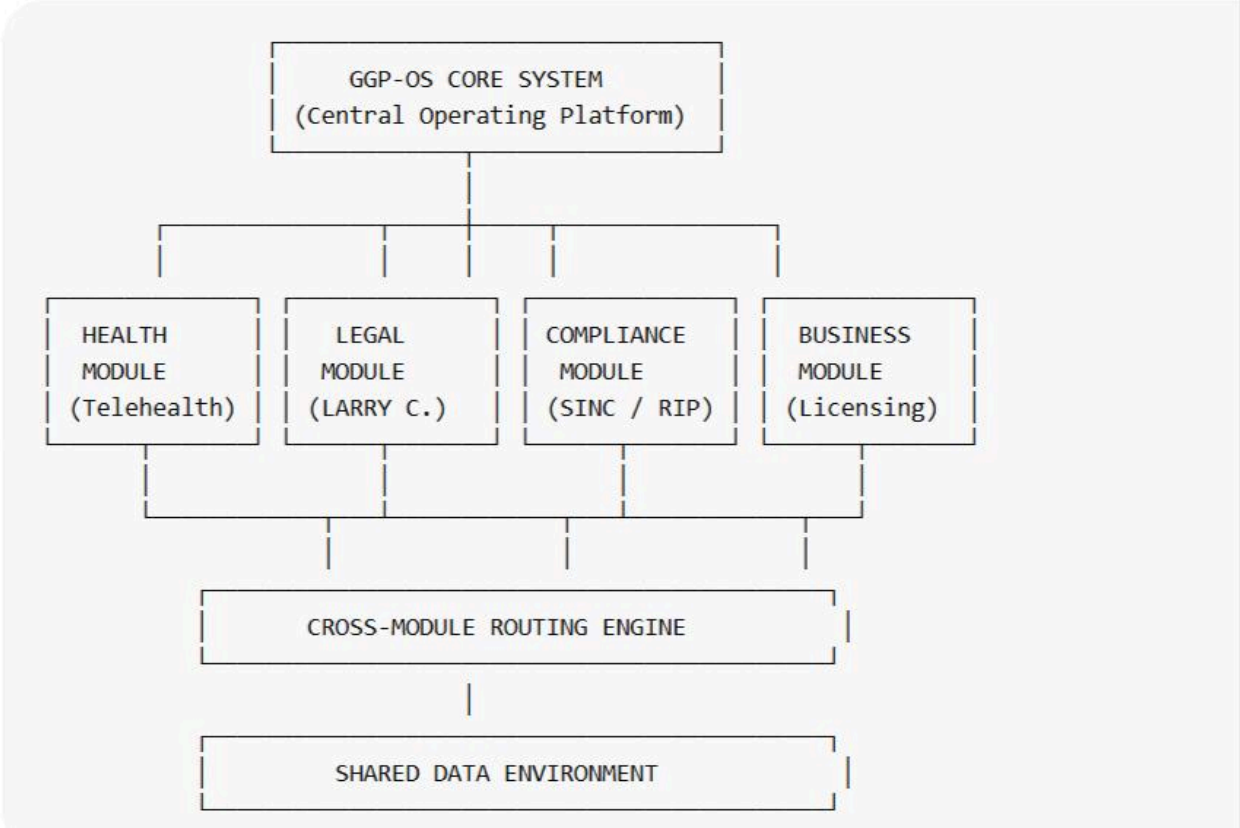
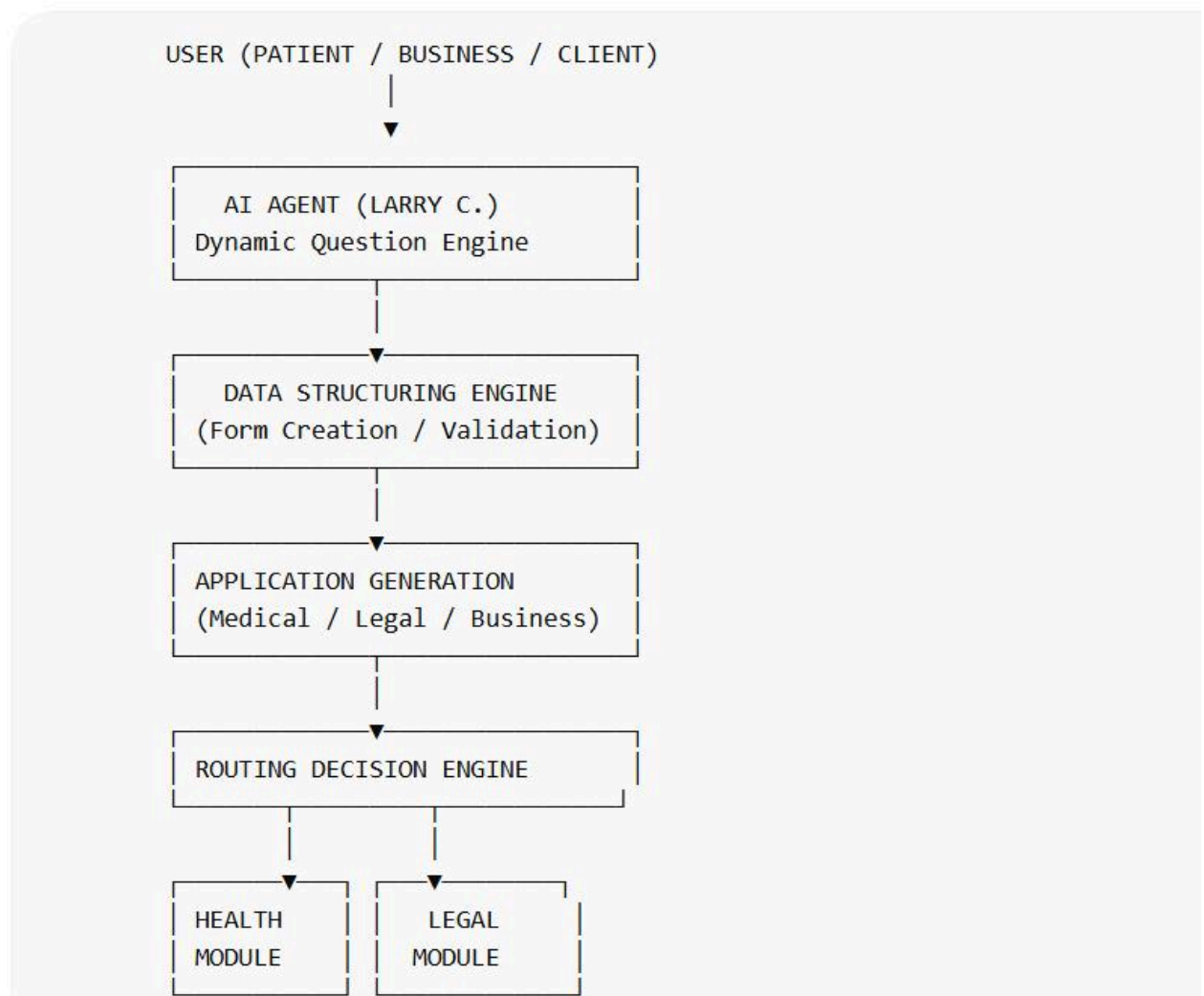


FIGURE 2 — AI INTAKE + APPLICATION FLOW

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**FIGURE 3 — CROSS-MODULE
ROUTING LOGIC**

FIGURE 3 — CROSS-MODULE ROUTING LOGIC

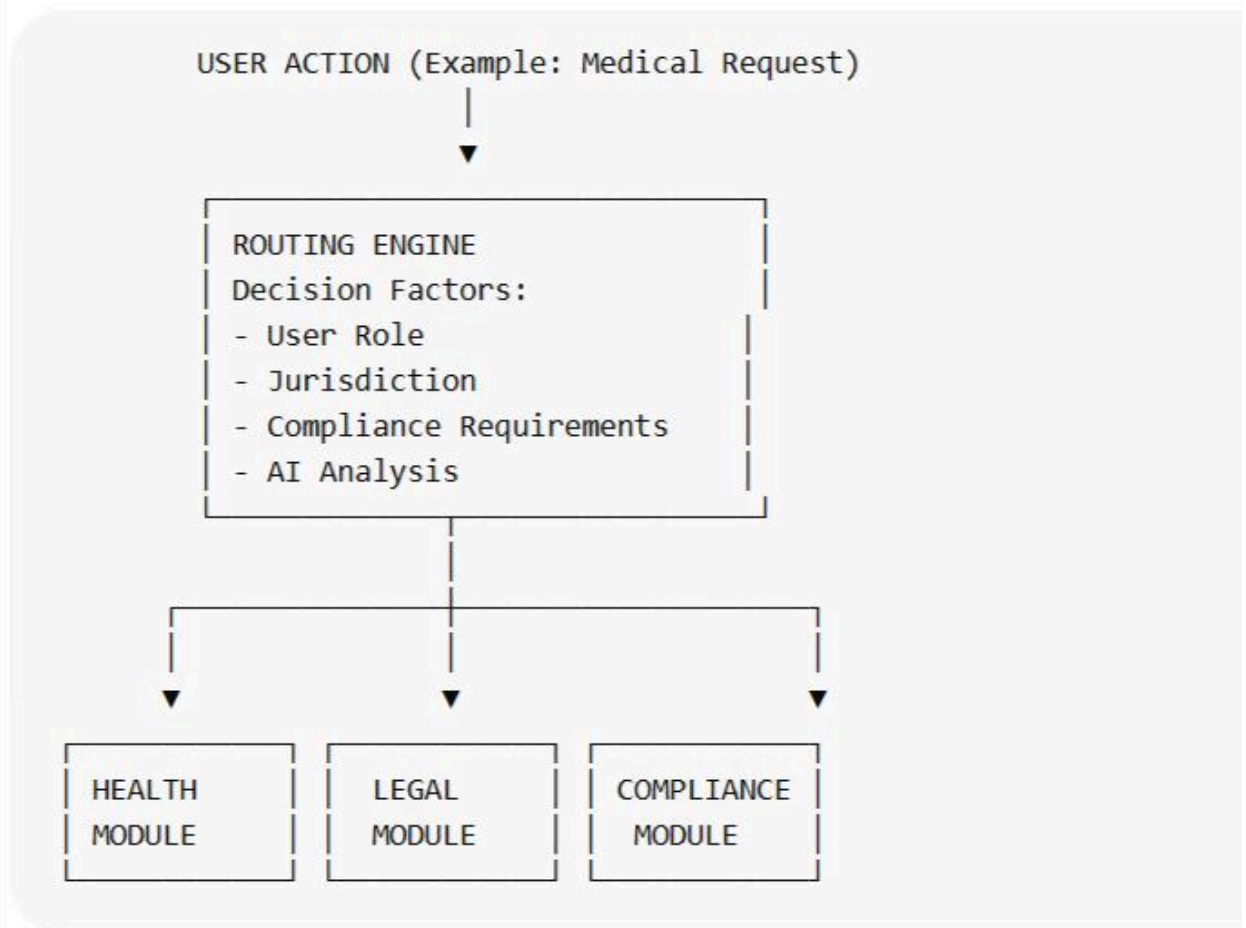


FIGURE 4 — RBAC (ROLE-BASED ACCESS CONTROL)

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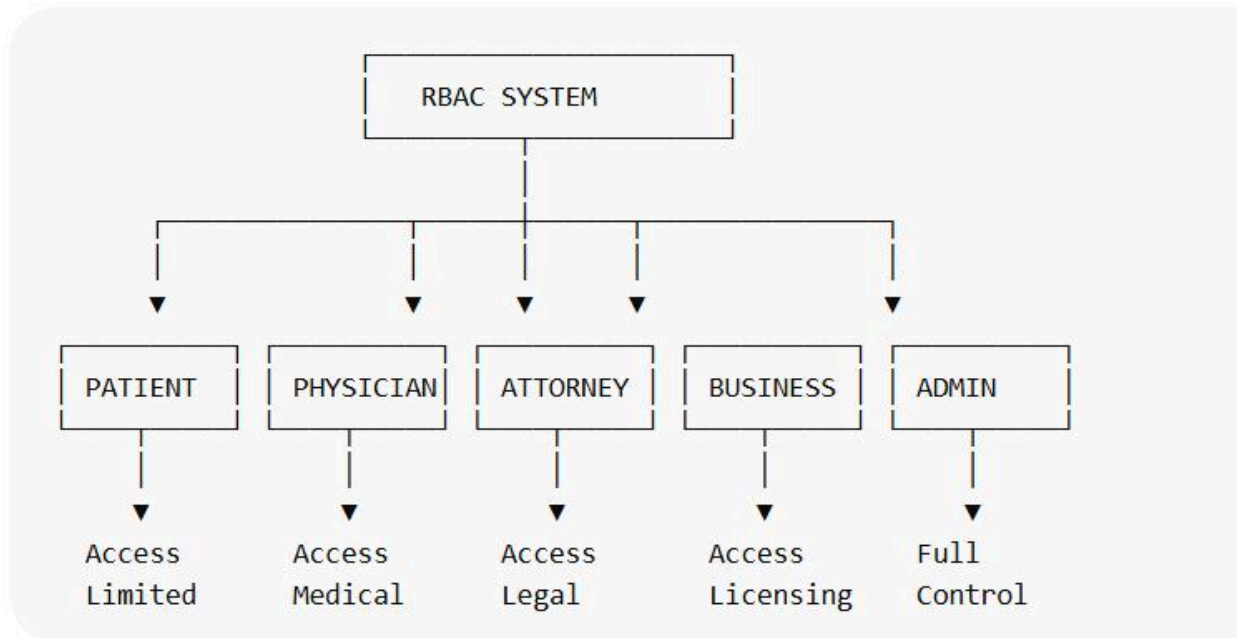


FIGURE 5 — TELEHEALTH + LEGAL + BUSINESS FLOW

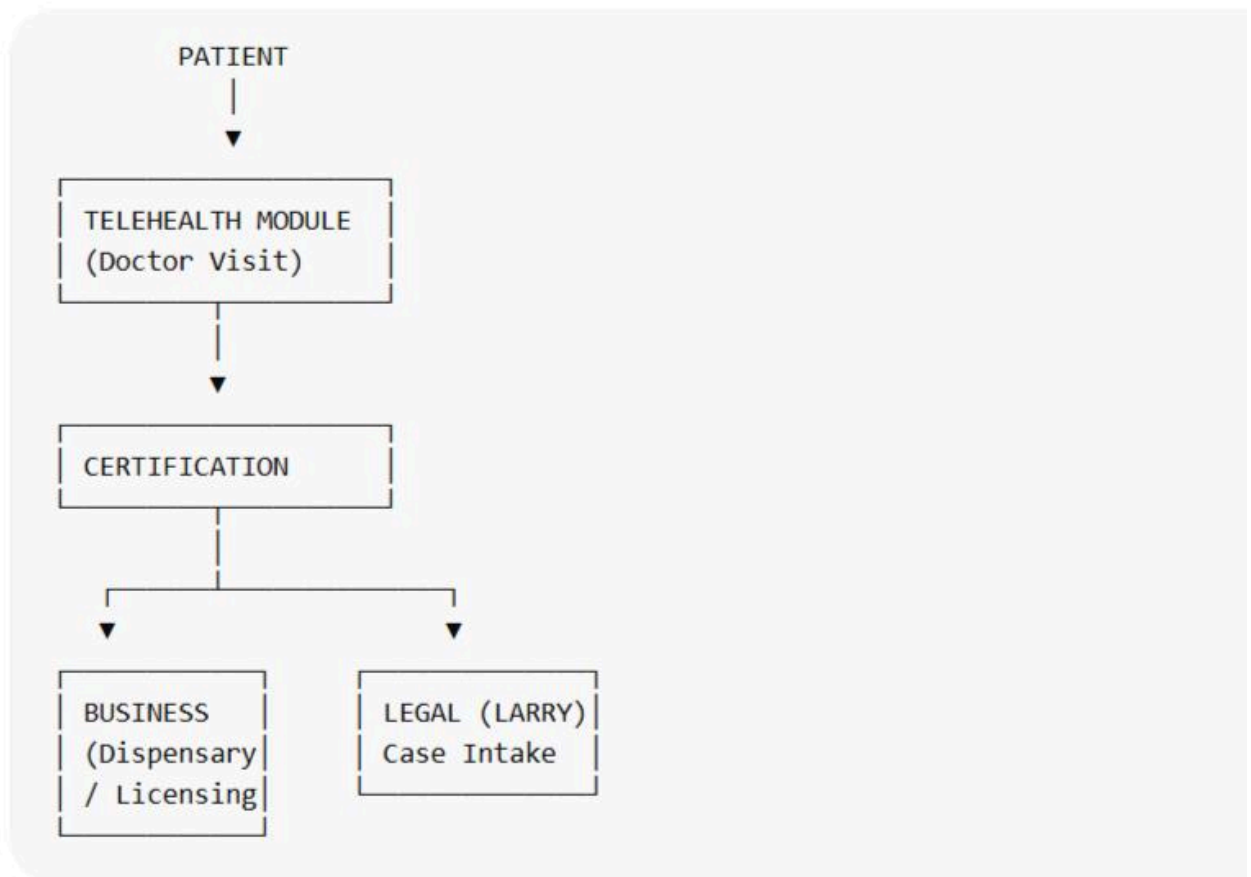
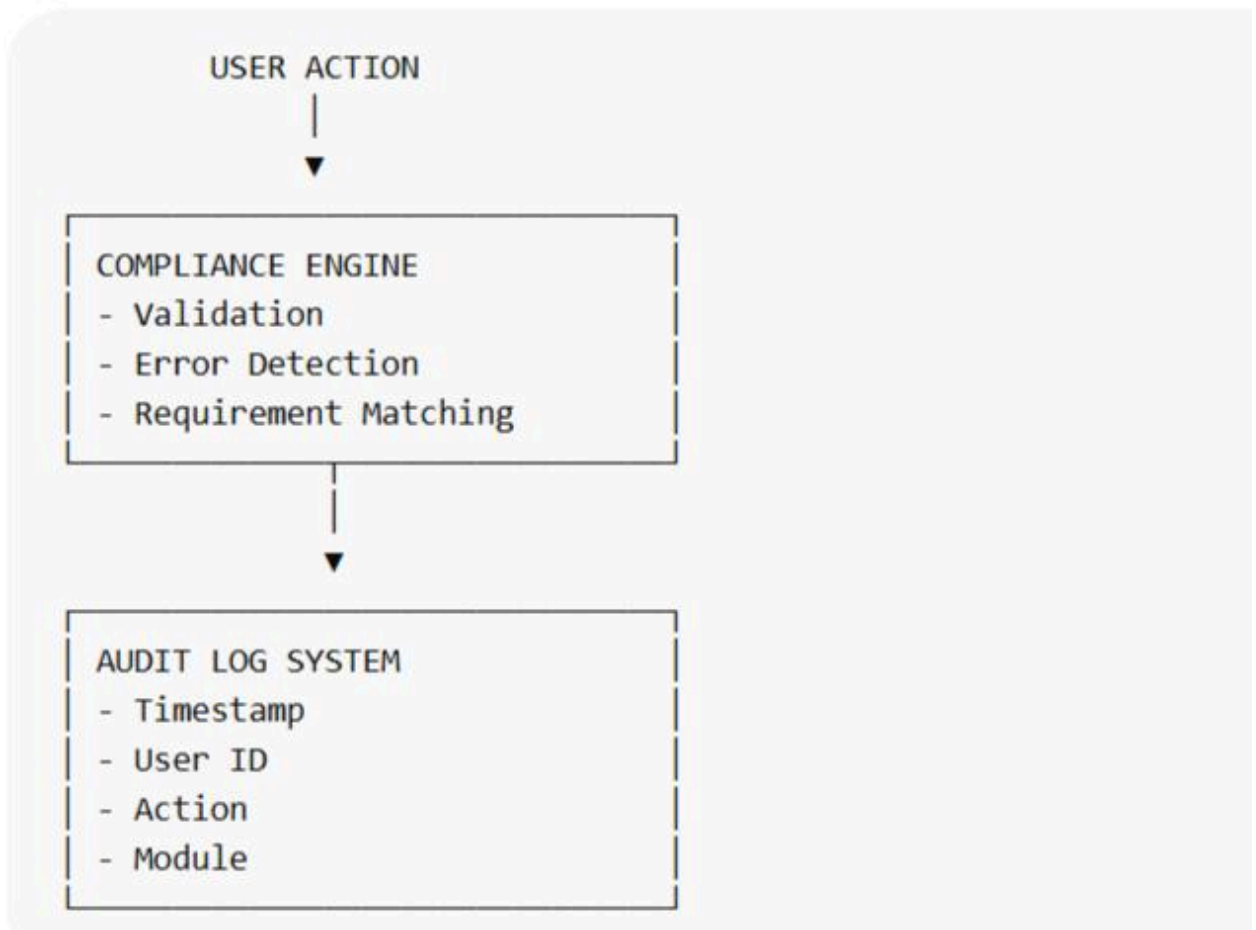


FIGURE 6 — COMPLIANCE + AUDIT SYSTEM



“The diagrams illustrated herein are exemplary representations and are not intended to limit the scope of the invention. Variations, modifications, and alternative configurations are within the scope of the described system.”